BETTER TOMORROW

Practise Problems - I Year

1. Check if a number is positive, negative, or zero.
2. Max of two numbers, smallest of two numbers.
3. Find the largest of three numbers.
4. Check if a number is even or odd.
5. Categorize a student's grade based on their marks (A, B, C, D, F).
6. Check if a person is eligible to vote.
7. Check if a number is divisible by 5.
8. Check if a number lies between two given numbers (range).
9. Check if the given char is a lower-case alphabet.
10. Check if a year is a leap year.
11. Determine if a character is a vowel or consonant.
12. Check if a number is divisible by 5 and 11.
13. Check if a number is divisible by 3 or 13.
14. Determine if a number is divisible by either 2 or 5 but not both.
15. Check if a character is uppercase or lowercase.
16. Check if a character is a digit or not.
17. Write a program to find the day of the week based on the number input (1-7).
18. Find the greatest of four numbers.
19. Check if a number is a perfect square or not.
20. Calculate the sum, difference, product of two numbers.
21. Find the quotient and remainder of two numbers using the modulo operator.
22. Area of a rectangle.
23. Area of a circle.
24. Use increment and decrement operators to modify values.
25. Use the sizeof operator to determine the size of variables.
26. Write a program to check if two numbers are equal, greater, or less than.
27. Swapping two variables using another variable.
28. Swap two numbers without using a temporary variable (using +, - operator).
29. Convert temperature from Celsius to Fahrenheit. (1°**C** × 9/5) + 32 = 33.8°**F**
30. Write a program to find the roots of a quadratic equation.
31. Convert a given number of seconds into hours, minutes, and seconds.

32. Write a program to calculate simple interest (p = principal, n = time, r = rate).
33. Write a program to calculate compound interest.
34. Write a program to swap two numbers using XOR.
35. Find the average of three numbers using arithmetic operators.
36. Write a program to check if a number is a power of 2.
37. Print the first n natural numbers.
38. Print countdown from a given n (e.g., 10, 9, 8… 3, 2, 1).
39. Find the sum of the first n natural numbers.
40. Find the factorial of a number.
41. Print the multiplication table of a given number.
42. Print all even numbers up to n / print all odd numbers up to n.
43. Count the number of odd/even numbers in a given range.
44. Find the sum of all odd/even numbers up to n.
45. Calculate the power of a number using a loop (without pow function).
46. Print the Fibonacci series up to n terms.
47. Print all numbers from 1 to n that are divisible by a given number.
48. Print all numbers divisible by 3 and 5 up to n.
49. In a range of 1 to n, print “fizz” if divisible by 3 and “buzz” if divisible by 5, and “fizz-buzz” if divisible by both.
50. Check if a number is prime.
51. Find the sum of squares of first n natural numbers.
52. Check if a number is an Armstrong number.
53. Print the ASCII value of all characters from A to Z.
54. Write a program to check if a number is strong (sum of factorials of digits equals the number).
55. Write a program to find the sum of all prime numbers up to n.
56. Print all perfect numbers up to n.
57. Print the cube of numbers from 1 to n.
58. Print numbers from 1 to 100 using a while loop.
59. Count the number of digits in an integer.
60. Find the sum of digits of a number using a while loop.
61. Write a while loop to print all digits of a number.
62. Write a program to reverse a number using a while loop.
63. Check if a number is a palindrome using a while loop.
64. Write a while loop to find the largest digit in a number.
65. Find the smallest digit in a number using a while loop.
66. Find the product of digits of a number using a while loop.

67. Write a while loop to print all factors of a number.
68. Write a while loop to calculate the sum of odd digits in a number.
69. Print the first n perfect numbers using a while loop.
70. Write a while loop to reverse the digits of a given number.
71. Store n numbers in an array and print them.
72. Find the largest element in an array.
73. Find the smallest element in an array.
74. Find the sum of elements in an array.
75. Find the average of elements in an array.
76. Reverse an array using swapping.
77. Write a program to search for an element in an array.
78. Find the sum of odd elements in an array.
79. Write a program to count the number of even and odd elements in an array.
80. Find the difference between the maximum and minimum elements in an array.
81. Write a program to find the second largest element in an array.
82. Write a program to find the second smallest element in an array.
83. Check if an array is sorted.
84. Write a program to count the occurrences of an element in an array.
85. Find the median of an array.
86. Write a program to check if two arrays are equal.
87. Find the frequency of each element in an array.
88. Write a program to check if an array is a palindrome.
89. Write a program to find the missing number in an array of 1 to n.
90. Find the product of all elements in an array.
91. Write a program to check if two arrays are reversed from each other.
92. Find the common elements between two arrays.
93. Find the majority element in an array.
94. Write a program to count the number of negative elements in an array.
95. Find the sum of prime elements in an array.
96. Write a program to find the second most frequent element in an array.
97. Write a program to find the length of a string.
98. Write a program to reverse a string.
99. Write a program to check if a string is a palindrome.
100. Write a program to concatenate two strings
101. Write a program to count the number of vowels in a string.
102. Write a program to remove all vowels from a string.

103. Write a program to find the frequency of each character in a string.
104. Write a program to check if two strings are anagrams.
105. Write a program to find the first non-repeating character in a string.
106. Write a program to replace all spaces in a string with a hyphen.
107. Write a program to create a function to find the maximum of two numbers.
108. Write a program to create a function to find the factorial of a number.
109. Write a program to create a function to check if a number is prime.
110. Write a program to create a function to calculate the sum of digits of a number.
111. Write a program to create a function to swap two numbers.
112. Write a program to create a function that returns the nth Fibonacci number.
113. Write a program to create a function that checks if a string is a palindrome.
114. Write a program to demonstrate pointer arithmetic.
115. Write a program to swap two numbers using pointers.
116. Write a program to find the sum of elements in an array using pointers.
117. Write a program to reverse a string using pointers.
118. Write a program to find the length of a string using pointers
119. Write a program to find the factorial of a number using recursion.
120. Write a program to find the nth Fibonacci number using recursion.
121. Write a program to find the greatest common divisor (GCD) of two numbers using recursion.
122. Write a program to reverse a string using recursion.
123. Write a program to calculate the power of a number using recursion